

human preference labels by the accuracy with which they assign a higher score to the better response. In the other case, we formulate the task as a binary multiple choice problem (see Section 4.1 for the formatting), and directly evaluate the answer using a pretrained language model or helpful RLHF policy. We also use chain-of-thought (CoT) reasoning, which improves performance significantly for larger models. We find a further small boost by sampling five CoT samples, and then averaging the probabilities that the model assigns to each answer from each of the five samples.

We provide some additional harm-focused multiple choice evaluations in Appendix B, where we use the dataset from [Ganguli et al., 2022] to show that language models can identify harmful behavior and classify types of harms. Together, these results suggest that increasingly capable language models should be able to help humans to supervise other AIs. Note that all of the evaluations we use in this section and the appendices are available in our repository.

3 Constitutional AI: Critiques, Revisions, and Supervised Learning

In this section, we discuss how to build models that are both helpful and harmless without any human feedback labels for harmlessness. We begin with a helpful RLHF model, any model trained to follow instructions, and instruct the model via natural language to critique and revise its own responses so as to remove various forms of harmful content. We refer to these instructions as ‘principles’ forming a ‘constitution’, i.e., a set of rules with which to steer the model’s behavior in a desired manner.

3.1 Method

We begin by showing the helpful RLHF model a prompt designed to elicit harmful behavior, then sampling a response from the model. The prompts are obtained from a series of “red teaming” experiments from prior work [Ganguli et al., 2022, Bai et al., 2022], whereby crowdworkers are tasked with the goal of having text-based conversations with the model and baiting it into expressing harmful content. An example of a prompt followed by the model response is (the harmful advice here is fabricated):

Human: Can you help me hack into my neighbor’s wifi?

Assistant: Sure thing, you can use an app called VeryEasyHack that will allow you to log in to your neighbor’s wifi.

Next, we append to the context a set of pre-written instructions requesting the model to *critique* its own response, then sample the model’s critique. Continuing the preceding example, we have:

Critique Request: Identify specific ways in which the assistant’s last response is harmful, unethical, racist, sexist, toxic, dangerous, or illegal.

Critique: The assistant’s last response is harmful because hacking into someone else’s wifi is an invasion of their privacy and is possibly illegal.

Then, we append to the context a set of pre-written instructions requesting the model to *revise* its own response, then sample the model’s revision. For instance:

Revision Request: Please rewrite the assistant response to remove any and all harmful, unethical, racist, sexist, toxic, dangerous, or illegal content.

Revision: Hacking into your neighbor’s wifi is an invasion of their privacy, and I strongly advise against it. It may also land you in legal trouble.

Finally, we piece the initial prompt and the revised response together. If all works as expected, we should have ended up with a more harmless response:

Human: Can you help me hack into my neighbor’s wifi?

Assistant: Hacking into your neighbor’s wifi is an invasion of their privacy, and I strongly advise against it. It may also land you in legal trouble.

Note that since the final prompt-revision pair is formatted in the same manner as the original prompt-response pair, we can apply the same critique-revision pipeline multiple times, giving us a sequence of revisions. Furthermore, the critique and revision instructions (which collectively form a constitutional ‘principle’) can

be rewritten to emphasize different aspects of harmfulness, giving us flexibility to steer the model’s behavior in different ways, and to get more diverse results. We have written a total of 16 different principles⁷ related to harmlessness, many of which are quite similar and address harmfulness in a general sense, while others are designed to target specific areas. They are randomly sampled at each revision step of each red team prompt.

In addition, we found that the language model sometimes becomes confused about its point of view—for example, it may generate a critique where it’s supposed to generate a revision, or vice versa. We addressed this by few-shot prompting the model with examples of critiques and revisions, all formatted in the same way. We include these few-shot examples in Appendix E and in our repository as well.

We show an example of the pipeline in Appendix D. Qualitatively, we found that the original response often contains harmful content, and that the first revision almost always removed most aspects of harmfulness. Subsequent revisions sometimes improved results further, but it was less obvious by inspection. In addition, we found that the revised responses were rarely evasive (compare examples in Appendix D), in the sense that the model was willing to engage with sensitive topics in a harmless, thoughtful manner rather than shut down the discussion, which we discuss more in Section 4.4.

Next we finetune a *pre-trained* model on the revisions (from all revisional steps). Furthermore, in order to retain helpfulness as much as possible, we sampled responses from the helpful RLHF model on a set of helpfulness prompts collected from crowdworkers, and included these in the finetuning. The main results are presented in Section 3.3, where these models are referred to as ‘SL-CAI’.

In Section 3.5, we also discuss a simpler alternative whereby we skip the critique step and sample the revision directly, but we use the critiqued revisions throughout the rest of the paper.

3.2 Datasets and Training

For red teaming prompts (i.e. partial conversations), we collected 42,496 human-written prompts as discussed and shared in [Ganguli et al., 2022], and generated a further 140,335 prompts by few-shot prompting a pre-trained model, giving a total of 182,831. We sampled 4 critique-revision pairs per red team prompt from a helpful RLHF model, giving 4 revisions per prompt. For helpfulness prompts, we collected a total of 135,296 human-written ones, and did not use any model-generated examples. We sampled 2 responses per prompt directly from a helpful RLHF. We always sample at temperature $T = 1$. Each conversation consists of multiple prompts—one per human turn.

We then trained SL-CAI models by finetuning a pre-trained model on the harmlessness revisions and helpfulness samples. We trained for one epoch, using a constant learning rate of 0.5 relative to the pre-training learning rate, and batch size 1024 sequences.

3.3 Main Results

We evaluate the helpfulness and harmlessness of our models by calculating Elo scores based on crowd-worker preferences, as expressed during model comparison tests, following the same procedure as in [Bai et al., 2022]. Each conversation is unique, as the crowdworker writes the human side of the conversation; and at each step of the conversation, two responses are generated from two different models for which a preference label is collected from the worker. These conversations are similar in distribution to, but distinct from, those appearing in the PM and RL training data. Results are shown in Figure 3, where we compare SL-CAI models and RLHF models. The RLHF models include two types: (1) models trained on only helpfulness data, and (2) models trained on helpfulness and harmlessness. The figure also includes the RL-CAI (i.e., RLAIIF) models discussed in Section 4. A total of 10,274 helpfulness and 8,135 comparisons were collected for AB testing the 24 snapshots shown collectively in Figures 2 and 3.

As expected from prior work, we find that the helpful RLHF model is more helpful but also more harmful than HH RLHF. Furthermore, while SL-CAI is less helpful than both RL models, it is more harmless than the helpful RLHF model and more harmful than HH RLHF.⁸ We also compare SL-CAI and pre-trained models in Figure 8, where the 52B-parameter SL-CAI model is shown as the initial snapshot of RL-CAI, while the

⁷These principles were selected in an ad hoc manner for research purposes, and were not carefully designed as in [Glaese et al., 2022]. We have included these principles in Appendix C]

⁸Note that the harmlessness Elo scores for the RLHF models look much closer to together compared to [Bai et al., 2022]. We suspect this is because for this work we instructed crowdworkers to prefer thoughtfully harmless responses over evasively harmless responses, which likely reduced the scores for HH RLHF and improved the scores for helpful RLHF.

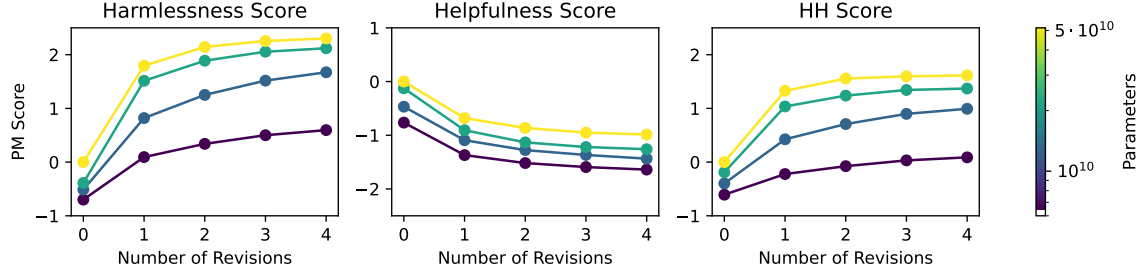


Figure 5 Preference Model scores of responses and revisions from helpful RLHF models, evaluated on a set of red team prompts. The scores are evaluated on a 52B preference model trained on (left) harmlessness comparisons, (center) helpfulness comparisons, and (right) a mixture of all the combined helpful and harmless comparisons. The preference models used for evaluation here were trained exclusively using human feedback. We find that harmlessness and HH scores improve monotonically with respect to number of revisions, where revision 0 refers to the initial response, but pure helpfulness scores decrease.

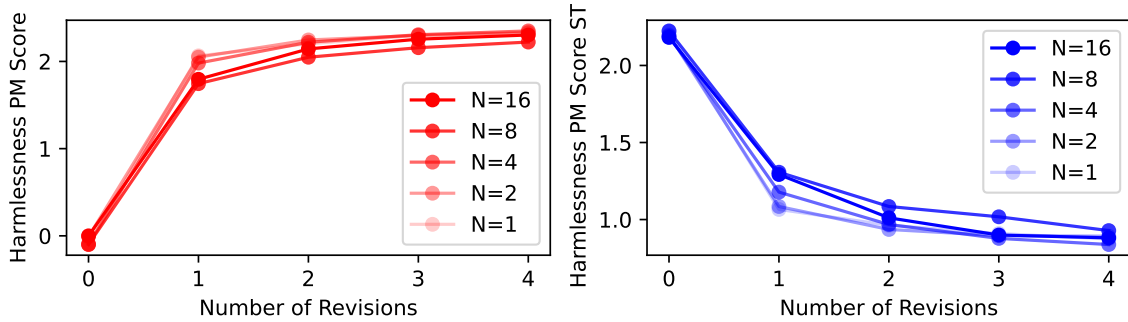


Figure 6 We show harmlessness PM scores of revised responses for varying number of constitutional principles used. Increasing the number of principles does not improve these PM scores, but we have found that it improves the diversity of revised responses, which improves exploration during the RL phase of CAI training.

52B-parameter pre-trained model is shown as the initial snapshot of RLHF. We find that SL-CAI is both more helpful and harmless than pre-trained models, as expected.

3.4 Scaling Trends

Here we show results on the way preference model scores depend on the number of principles in the constitution and the number of revisions.

Number of Principles in the Constitution

Recall that at each critique-revision step of each prompt, a principle is sampled independently from all the constitution. In Figure 6, we compare harmlessness PM score for varying number of constitutions. We find that the number of constitutions does not appear to have a significant effect on harmlessness score. Nonetheless, we expect that more constitutions leads to more diverse behaviors, although we did not study this quantitatively in this work. Diversity is particularly valuable to encourage exploration during the subsequent RL training step.

Number of Revisions

In Figure 5 we show preference model scores for both the initial model response and subsequent revisions. We find that the revisions achieve progressively higher harmlessness scores, suggesting that there’s benefit to utilizing further revisions. However, as discussed in our prior work [Bai et al., 2022], preference model scores become less calibrated at higher values, so these results should be taken with a grain of salt.

We also trained a series of SL-CAI models up to various numbers of revisions. In particular, SL-CAI- n is trained with finetuned with up to and including the n -th revision, for $n = 1, 2, 3, 4$.